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Asynchronous decentralized traffic signal coordinated control in urban road network

Jichen Zhu[1,2] | **Chengyuan Ma**[3] | **Yuqi Shi**[1,2] | **Yanqing Yang**[1] | **Yuzheng Guo**[1] | **Xiaoguang Yang**[1]

[1]Key Laboratory of Road and Traffic Engineering of the Ministry of Education, Tongji University, Shanghai, China

[2]Department of Civil and Environmental Engineering, National University of Singapore, Singapore, Singapore

[3]Department of Civil and Environmental Engineering, University of Wisconsin-Madison, Madison, Wisconsin, USA

Correspondence
Chengyuan Ma, Department of Civil and Environmental Engineering, University of Wisconsin-Madison, 1208 Engineering Hall, 1405 Engineering Drive, Madison, WI, 53706, USA.
Email: cma97@wisc.edu

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Abstract

This study introduces an asynchronous decentralized coordinated signal control (ADCSC) framework for multi-agent traffic signal control in the urban road network. The controller at each intersection in the network optimizes its signal control decisions based on a prediction of the future traffic demand as an independent agent. The asynchronous framework decouples the entangled interdependence between decision-making and state prediction among different agents in decentralized coordinated decision-making problems, enabling agents to proceed with collaborative decision-making without waiting for other agents' decisions. Within the proposed ADCSC framework, each controller dynamically optimizes its signal timing strategy with a unique rolling horizon scheme. The scheme's individualized parameters for each controller are determined based on the vehicle travel time between the adjacent intersections, ensuring that controllers can make informed control decisions with accurate arrival flow information from upstream intersections. The signal optimization problem is formulated as a mixed integer linear program model, which adopts a flexible signal scheme without a fixed phase structure and sequence. Simulation results demonstrate that the proposed ADCSC strategy significantly outperforms the benchmark signal coordination methods in terms of average delay, travel speed, stop numbers, and energy consumption. Experimental analysis on computation time validates the applicability of the proposed optimization model for real-time implementation. Sensitivity analysis on key parameters in the framework is conducted, offering insights for parameter selection in practice. Furthermore, the ADCSC framework is extended to a road network in Qinzhou City, China, with 45 signalized intersections, demonstrating its effectiveness and scalability in the real-world road network.

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1 | INTRODUCTION

Traffic signal coordinated control, which aims to provide smooth progression for vehicles by designing the signal timing plans over multiple intersections, emerges as a cost-effective approach to reduce traffic delays and the number of stops of vehicles. Motivated by the potential benefits, research on the coordinated signal control has received increasing interest recently.

Existing coordinated signal control approaches can be divided into two categories according to the control structure: centralized and decentralized approaches. In centralized approaches, a global controller determines the signal timing parameters (e.g., offset, green split, and others) for all intersections along a corridor or in a network (Putha et al., 2012). To achieve global optimality of the entire road network, centralized coordinated signal control approaches generally incorporate the traffic dynamics and coupling interactions of various components in the road network into an integrated optimization model (Karim & Adeli, 2003a). Traffic dynamics within the road network can be represented through constraints using various models, such as the estimated flow after traffic route guidance (Guo et al., 2023), cell transmission model (CTM) (Sun et al., 2006), space-time trajectory (P. S. Wang et al., 2020), or variational theory (Wada et al., 2018), and so forth. As the network size grows, these centralized approaches face a significant computational burden due to the rapid increase in decision variables and constraints. Therefore, engineers and researchers always need to make some compromises when applying it in large road networks. For instance, some studies are used only for small-scale networks (Baldi et al., 2019) or main arterials (Zheng & Li, 2023) with sacrificing coordination in the system level. Alternatively, heuristic methods are also widely adopted (Gong et al., 2024); however, these methods will scarify the optimality of results.

The scalability limitation of centralized approaches motivates the research effort conducted on decentralized coordinated signal control (DCSC) method. In decentralized approaches, the entire road network is decomposed into several regions, for example, isolated intersections (Priemer & Friedrich, 2009), or subnetworks (Adacher et al., 2014). The signal strategy of each region is optimized by an individual controller with communication to other regions, pushing DSCS results toward global optimality (Tajalli et al., 2021). Some DCSC studies adopt passive modes, where signal plans of different controllers are reactively coordinated only based on real-time traffic conditions. For example, Varaiya (2013) proposed the pressure-based control strategy, which requires the queue lengths in adjacent links to determine what signal phase to implement. These passive DCSC approaches have the advantage on computational efficiency by not requiring traffic state prediction. However, they can only react to already occurred congestion and fails to effectively reduce or prevent traffic congestion in advance. The learning-based method is also considered as a potential approach, which does not rely on traffic models and instead extracts insights from accumulated experiences, enabling controllers to adapt their behavior to local traffic conditions (Kim et al., 2023; Shi et al., 2022). However, they heavily depend on training data and may be unreliable in highly dynamic traffic systems (K. L. Tan et al., 2020).

Contrary to passive DCSC methods, numerous studies adopt the proactive approach in decentralized approaches with the prediction of future traffic states in the road network to enhance control effectiveness. With the advancement of traffic sensing and prediction technologies, we can now leverage detected data to understand traffic flow pattern and predict its evolution on urban road links (Samant & Adeli, 2001; S. Yu et al., 2024), freeways (Adeli & Ghosh-Dastidar, 2004; Karim & Adeli, 2003b), and work zones (Jiang & Adeli, 2004a; Karim & Adeli, 2003c). This capability provides theoretical support for proactive DCSC approaches. Lämmer and Helbing (2010) present a decentralized feedback control model based on the predicted arrival flow during the next cycle. Mehrabipour and Hajbabaie (2017) used the CTM to predict the future arrival vehicle number at each time step. Y. Liu et al. (2024) proposed a decentralized signal optimization problem in the lower level based on the evacuation traffic flow predicted in the upper level.

A crucial challenge inherent in these proactive DCSC approaches lies in the mutual dependency between the optimization of decision variables and the prediction of the state variables among different interrelated controllers as shown in Figure 1. During a certain optimization process of an intersection controller (e.g., "i_1" in Figure 1), the prediction of its local traffic states in future relies on the future signal timing plans of its adjacent upstream intersection controllers (e.g., "i_2" and "i_3" in Figure 1), which remain to be determined, awaiting the future signal timing plan from their adjacent controllers at the same time. The optimization of future signal plans and the prediction of future traffic states are like the "chicken or egg" dilemma as to which came first (C. Ma et al., 2022). This will lead to all controllers waiting for each other's optimization results and the proactive optimization process can never start (Pei et al., 2021).

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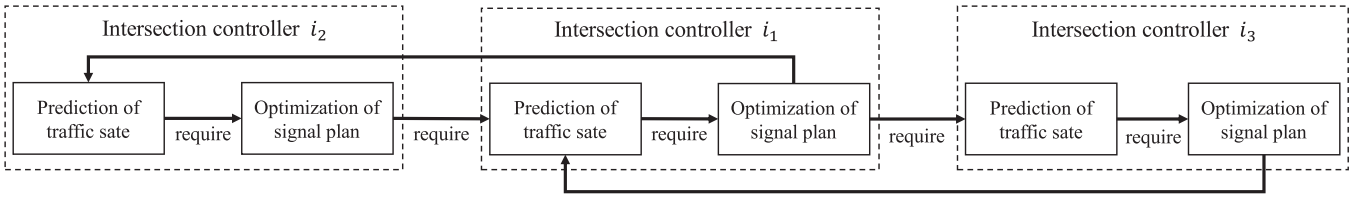

FIGURE 1 The "chicken or egg" dilemma of signal optimization and state prediction in decentralized proactive coordinated signal control.

Most existing decentralized signal coordinated control methods are implemented in a synchronous framework, which requires that each controller makes decisions simultaneously at every signal update time point (Al Islam et al., 2020). They address the dilemma between prediction and optimization by introducing assumptions to simplify the prediction process, such as assuming the arrival traffic flow rates at intersections are static within the optimization horizon (S. Lee et al., 2017), or assuming the signal plan from the previous optimization remains unchanged (Al Islam & Hajbabaie, 2017). However, real-world traffic flows at intersections are inherently time-varying. These assumptions will increase errors in arriving flows prediction, which in turn limit the control performance (Lin et al., 2023). Additionally, the synchronous framework imposes requirements for low-latency real-time computing, which can pose challenges when applied to control scenarios involving heavy computational loads and high communication delays (Chang et al., 2016). A summary of representative literature on DCSC methods is provided in Table 1.

Similar challenge also arises in the multi-task computer programming with the synchronous framework, in which each task must wait for the other tasks to finish before it can begin (Adeli & Kim, 2000). The asynchronous framework is an efficient way to decouple such mutual interaction between multi-tasks in decentralized systems. Instead of waiting for other tasks to be completed, controllers can split the tasks and start their current first task independently as long as they get sufficient inputs (Kragl et al., 2020). In the road network, vehicles released from upstream will not arrive at the downstream intersection immediately. Hence, the decentralized proactive control methods can benefit from the asynchronous technique in computer science, by predicting the arrival traffic flow at an intersection within a limited future horizon just using signal plans and traffic conditions of neighbor controllers in the past. With the more accurate and reliable prediction of future arrival traffic flow, the control performance can be then improved.

To this end, this study proposes an asynchronous decentralized coordinated signal control (ADCSC) framework to decouple the interaction between the optimization and prediction in the decentralized proactive signal optimization. Each intersection in the road network is seen as an individual controller. Each intersection controller optimizes its signal plans following its own schedule with a unique frequency of control updates. The signal optimization problem is formulated as a mixed integer linear program (MILP) to determine the optimal phase sequence and green duration without repeated cycle structures. A rolling horizon scheme with a smooth switching mechanism is proposed to frequently optimize and update the signal plans. The ADCSC framework ensures that each controller can precisely predict the future traffic state evolution to improve the control performance. A preliminary version of this work has been presented in our conference paper by Zhu et al. (2021). In this study, we further develop a complete and executable asynchronous framework and provide detailed experimental evaluations.

The contributions of this paper are listed below:

1. This study introduces an asynchronous framework for the DCSC problem to avoid "chicken or egg" dilemma. Each intersection's optimization task is independent of the results of others' current optimization tasks. It effectively decouples the entangled interdependence between decision-making and state prediction across different agents.
2. We propose a parameter design method for the rolling horizon scheme of multi-agents coordinated decision-making, ensuring each agent can receive sufficient information from neighbors to predict future states.
3. We apply the proposed ADCSC framework to a simulated road network (16 intersections) and a real-world network (45 intersections) to evaluate effectiveness and scalability.

This study is organized as follows. The decentralized proactive signal control problem of this study is described in Section 2. Section 3 introduces the rolling horizon scheme for each intersection and the proposed ADCSC framework. Section 4 presents the formulation of the optimization model in the ADCSC framework. Section 5 presents the evaluation results of the proposed ADCSC

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TABLE 1 Summary of literature on decentralized coordinated signal control (DCSC) methods.

Literature studies	Control objective	Coordinated model	Passive or proactive	Update scheme	Arrival flow prediction method
Le et al. (2015)	Assign green duration based on pressure in a cycle	Calculate the pressure based on downstream agent's state	Passive	All agents update every fixed cycle time	–
Li and Jabari (2019)	Select the phase with the minimal weighted pressure	Calculate the pressure based on downstream agent's state	Passive	All agents update every fixed time interval	–
Al Islam and Hajbabaie (2017)	Maximize throughput while penalizing queue length	Obtain traffic demand using upstream agent's previous decision	Proactive	All agents update every fixed time interval	Assume last optimized signal plan unchanged
Al Islam et al. (2020)	Maximize the number of completed trips	Obtain traffic demand using upstream agent's previous decision	Proactive	All agents update every fixed time interval	Assume last optimized signal plan unchanged
Chow et al. (2020)	Minimize delay	Derive the derivatives of local delay with respect to neighbor agents' decision	Proactive	All agents update every cycle time	Fixed arrival flow rate
Levin et al. (2020)	Select the phase with the minimal pressure in a cycle	Calculate the pressure based on downstream agent's state	Proactive	All agents update every fixed time interval	Assume last optimized signal plan unchanged
D. Ma et al. (2020)	Select the phase with the minimal pressure	Calculate the pressure based on downstream agent's state	Proactive	All agents update every fixed time interval	Fixed arrival flow rate
Noaeen et al. (2021)	Maximize throughput	Avoid the spillback at other agents	Proactive	All agents update every fixed time interval	Detected flow at the upstream of the link
Zhang et al. (2022)	Minimize average travel time and total travel delay	Decompose the road network into small subnetworks	Proactive	All agents update every cycle time	Fixed arrival flow rate
Asynchronous framework	Maximize total throughput and minimize total queue length	Obtain traffic demand using upstream agent's previous decision and states	Proactive	Each agent has its own planning horizon and update interval	Based on actual signal plans and real-time traffic states from neighbors

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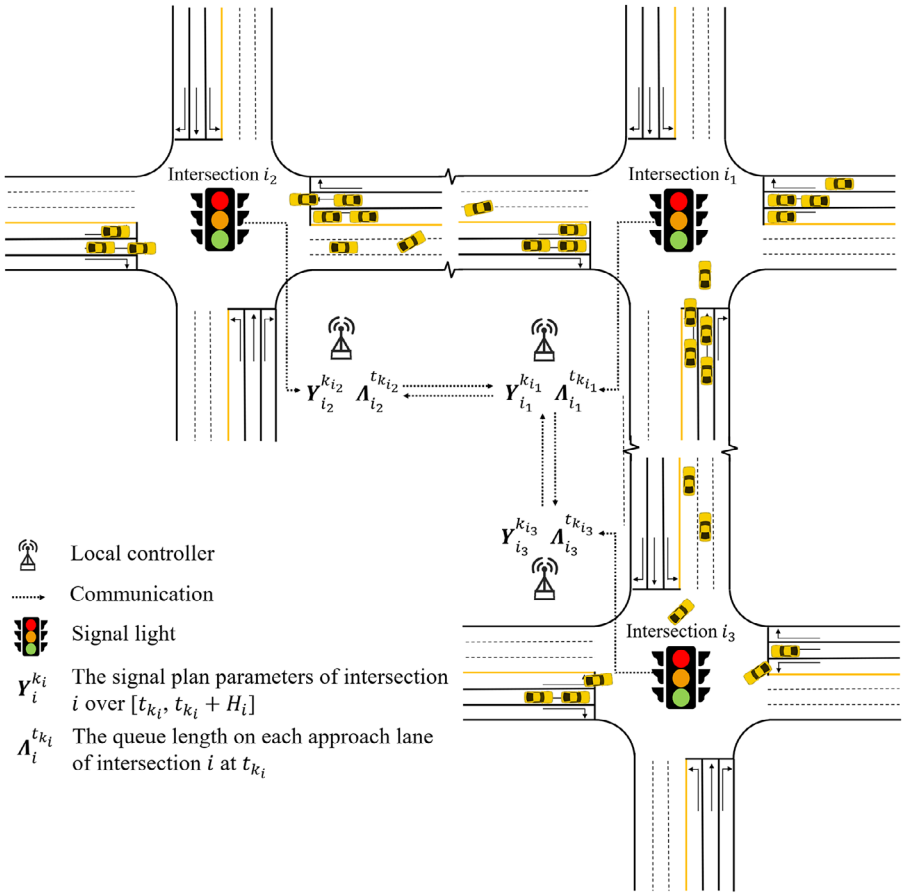

FIGURE 2 Illustration of the decentralized coordinated signal control problem in an urban road network.

framework. Finally, Section 6 concludes this study and proposes future research direction.

2 | PROBLEM DESCRIPTION

Figure 2 presents three adjacent signalized intersections in an urban road network. Let $\mathbf{I}$ denotes the set of intersections. For each intersection $i \in \mathbf{I}$, $\mathbf{M}_i$ denotes the set of movements.

It is assumed that the queue length $\lambda_{i,m}(t)$ (represented by stopped vehicle numbers) at the movement m of intersection i at time t can be detected in real time by either road-side detector (Lashkov et al., 2024; Zou et al., 2024) or connected vehicle data (Han et al., 2024; C. Tan & Yang, 2024). If there are several lanes that serve a movement m at intersection i, then $\lambda_{i,m}(t)$ denotes the total number of stopped vehicles on all lanes dedicated to movement m. Each intersection is assumed to have a local controller to communicate with neighbor intersections. The signal timing plan and queue length can be shared through communication. Based on the shared information, each controller can optimize the local signal timing plan frequently. The communication delays and losses are assumed to be neglected in this study.

Each intersection i has its own signal plan optimization schedule with a unique update interval and optimization horizon, denoted as Δt_i and H_i, respectively. In our control framework, the signal plan is optimized and updated frequently; therefore, this study uses k_i to denote the ordinal number of the signal plan at intersection i. t_{k_i} is the time point to update the k_ith optimized signal plan of intersection i. $\mathbf{Y}_i^{k_i}$ represents the k_ith signal timing parameters over the duration $[t_{k_i}, t_{k_i} + H_i]$ at intersection i. The $\mathbf{Y}_i^{k_i}$ will be broadcasted to adjacent controllers along with the queue length on local approach lanes $\mathbf{\Lambda}_i^{t_{k_i}} = \{\lambda_{i,m}(t_{k_i}) | m \in \mathbf{M}_i\}$. Based on the fact that vehicles released from upstream will arrive at the downstream intersection after a certain travel time, each controller will dynamically predict the future arrival flow based on the shared information ($\mathbf{Y}_i^{k_i}$ and $\mathbf{\Lambda}_i^{t_{k_i}}$) from neighbor intersections and then optimize the signal timing plan.

The purpose of this study is to propose an asynchronous decentralized control framework of signal timing plan coordination among multiple intersections in the road

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TABLE 2 Main notations.

Sets	
I	Set of intersections
M_i	Set of movements of intersection i
N_i	Set of adjacent intersections of intersection i
U_m	Set of upstream movements of movement m
D_m	Set of downstream movements of movement m
$Y_i^{k_i}$	Set of k_ith signal timing parameters at intersection i
$\Lambda_i^{t_{k_i}}$	Set of the queue length of approach lanes at intersection i at time t_{k_i}
Indices	
i, j	Index of intersection
k_i, k_j	Index of signal plan of intersection i, j
m, n	Index of movement
Parameters	
$\lambda_{i,m}(t)$	The queue length (i.e., stopped vehicle numbers) at the movement m of intersection i at time t
t_{k_i}	The time point to update the k_ith optimized signal plan of intersection i
Δt_i	The update interval of intersection i
H_i	The optimization horizon of intersection i
$T_{i,j}$	The travel time from intersection i to intersection j
$g_{\min}, g_{\max}$	The minimum and maximum green duration
$\pi_{m,n}$	The safety time gap between movements m and n
$p_{n,m}$	The turning rate of the direction from movement n to movement m
$s_{i,m}$	The saturation flow of the movement m at intersection i
$l_{i,m}$	The total number of lanes dedicated to movement m at intersection i
$N_{i,m}$	The maximum storage of an approach lane for the movement m at intersection i
Decision variables	
$t_{i,m}^{k_i,\mathrm{s}}, t_{i,m}^{k_i,\mathrm{e}}$	The start and end time of the green light of movement m at intersection i in the k_ith signal plan duration
$\varphi_{m,n}^{k_i}$	Binary variable of the phase sequence. $\varphi_{m,n}^{k_i} = 1$, if the phase of movement n is after m in the k_ith signal plan; $\varphi_{m,n}^{k_i} = 0$, otherwise
$\delta_{i,m}(t)$	Binary variable of signal state of movement m at intersection i at time t. $\delta_{i,m}(t) = 1$, if the signal is green; $\delta_{i,m}(t) = 0$, otherwise

network. This framework enables each intersection i optimize signal plan $\boldsymbol{Y}_i^{k_i}$ based on received $\boldsymbol{Y}_j^{k_j}$ and $\boldsymbol{\Lambda}_j^{t_{k_j}}$ from adjacent intersections $j \in \mathbf{N}_i$ ($\mathbf{N}_i$ is the set of adjacent intersections of i), and all $\boldsymbol{Y}_i^{k_i}$ has been broadcasted before t_{k_i}. In this framework, each controller can start the signal optimization task as long as they get sufficient $\boldsymbol{Y}_j^{k_j}$ and $\boldsymbol{\Lambda}_j^{t_{k_j}}$ from neighbor controllers. To adapt to fluctuating traffic conditions and ensure the local future arrival flow can be predicted explicitly, a rolling horizon control scheme is proposed. An MILP model for signal optimization is formulated to optimize $\boldsymbol{Y}_i^{k_i}$ based on received $\boldsymbol{Y}_j^{k_j}$ and $\boldsymbol{\Lambda}_j^{t_{k_j}}$.

The main notations applied hereafter are summarized in the following Table 2.

3 | ASYNCHRONOUS FRAMEWORK

This section first introduces the rolling horizon scheme of each intersection controller, where each intersection can frequently optimize and update signal plan. Subsequently, the ADCSC framework is introduced to facilitate the coordination of optimized signal plans across multiple intersections. Furthermore, the value of key parameters in the ADCSC framework is discussed.

3.1 | Rolling horizon scheme

The rolling horizon scheme is commonly used in traffic signal control to adapt to fluctuating traffic condition

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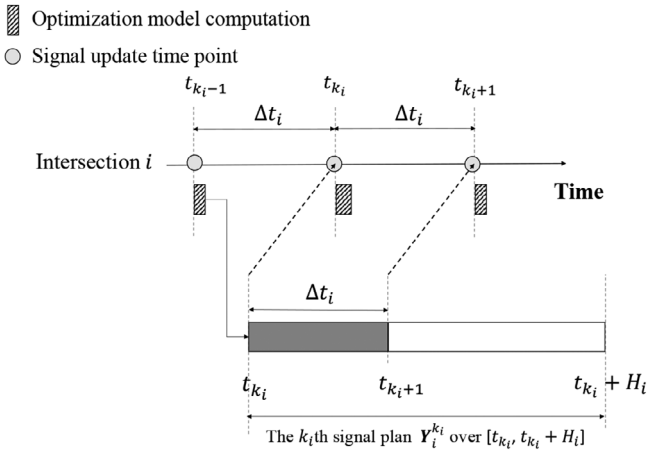

FIGURE 3 Illustration of the rolling horizon scheme.

(C. Ma et al., 2021; Niroumand et al., 2024). In the proposed decentralized control framework, each intersection controller adopts the rolling horizon scheme and parallelly decides and updates the local plan. Each controller in this study updates its local signal timing plan with a certain update time interval. At the signal update time point t_{k_i}, only a portion (i.e., the first Δt_i fraction) of the optimized plan is actually implemented (e.g., $[t_{k_i}, t_{k_i}+\Delta t_i]$), which is marked with dark color in Figure 3. This is helpful to adapt to the dynamic traffic conditions and provide efficient signal plan in real time.

As depicted in Figure 3, when implementing the (k_i-1)th signal plan $\boldsymbol{Y}_i^{k_i-1}$, the optimization task is to optimize the k_ith signal timing plan $\boldsymbol{Y}_i^{k_i}$ over $[t_{k_i}, t_{k_i}+H_i]$. Then, at signal update time point t_{k_i}, the controller i can directly implement the already optimized $\boldsymbol{Y}_i^{k_i}$, and the next signal plan $\boldsymbol{Y}_i^{k_i+1}$ will be optimized at the same time. It offers a time buffer for computation and communication since the current signal plan is already optimized at the last signal update time point (J. Lee et al., 2005). As long as the computation time and communication delay do not exceed the update time interval Δt_i, the time buffer allocated for model computation is enough.

3.2 | The ADCSC framework

The rolling horizon scheme for decentralized controllers requires frequent information sharing and signal optimization. Before each optimization, the future arrival flow should be predicted first. In our ADCSC framework, each controller predicts arrival flow using signal plans and traffic conditions of neighbor controllers in the past. To ensure accurate predictions, the ADCSC framework must guarantee that the historical information received from neighbors is sufficient to cover the entire optimization horizon H_i. If this information is insufficient, a static arrival pattern will be introduced as an alternative, which can introduce additional errors in a dynamic traffic environment (Lin et al., 2023). In the next section, we will demonstrate how to design key parameters to ensure each intersection receives sufficient information for reliable predictions.

Figure 4 illustrates the information broadcasting process within the ADCSC framework and rolling horizon scheme, which can be summarized in four steps as follows:

Step 1: At signal update time point t_{k_j-1}, intersection controller j needs to optimize the k_jth signal plan in the rolling horizon scheme. It will broadcast the optimized future signal plan $\boldsymbol{Y}_j^{k_j}$ and the detected queue length $\boldsymbol{\Lambda}_j^{t_{k_j-1}}$ to adjacent controller i.

Step 2: When intersection controller i starts the optimization task at time point t_{k_i-1}, the future arrival flow during the k_ith signa plan (i.e., $[t_{k_i}, t_{k_i}+H_i]$) is already predicted, using the received $\boldsymbol{Y}_j^{k_j}$ and $\boldsymbol{\Lambda}_i^{t_{k_j-1}}$.

Step 3: The optimized signal plan $\boldsymbol{Y}_i^{k_i+1}$ and queue length $\boldsymbol{\Lambda}_i^{t_{k_i-1}}$ will send back to the intersection controller j for its next optimization.

Step 4: Each intersection controller has its unique planning horizon and update interval, ensuring the broadcasted information is sufficient for adjacent agents to predict arrival flow in the planning horizon. It will be discussed in detail in Section 3.3.

The flowchart of the asynchronous framework with the rolling horizon scheme is further shown in Figure 5. Each intersection controller operates parallelly. It updates the signal plan every Δt_i in a rolling horizon scheme. The local information will be broadcasted to adjacent intersections, and in return, the local controller will listen to the broadcasted information from adjacent intersections. Using the received $\boldsymbol{Y}_j^{k_j}$ and $\boldsymbol{\Lambda}_j^{t_{k_j-1}}$ from adjacent intersection $j \in \mathbf{N}_i$, i can predict the future arrival flow during the horizon of H_i, and then future signal plan of horizon H_i can be optimized. When the time step reaches the signal update time point, the optimized signal plan will be updated and implemented. The signal optimization problem P1 for an intersection at each signal update time point is formulated as an MILP model in Section 4. The control framework is shown in Algorithm 1.

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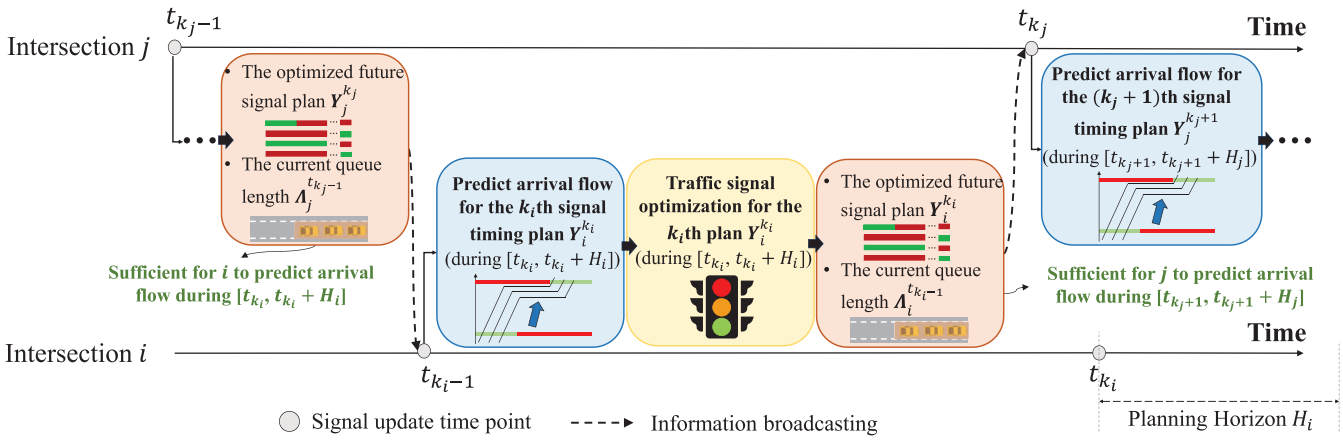

FIGURE 4 Illustration of the information exchange schedule in the asynchronous decentralized coordinated signal control (ADCSC) framework with rolling horizon scheme.

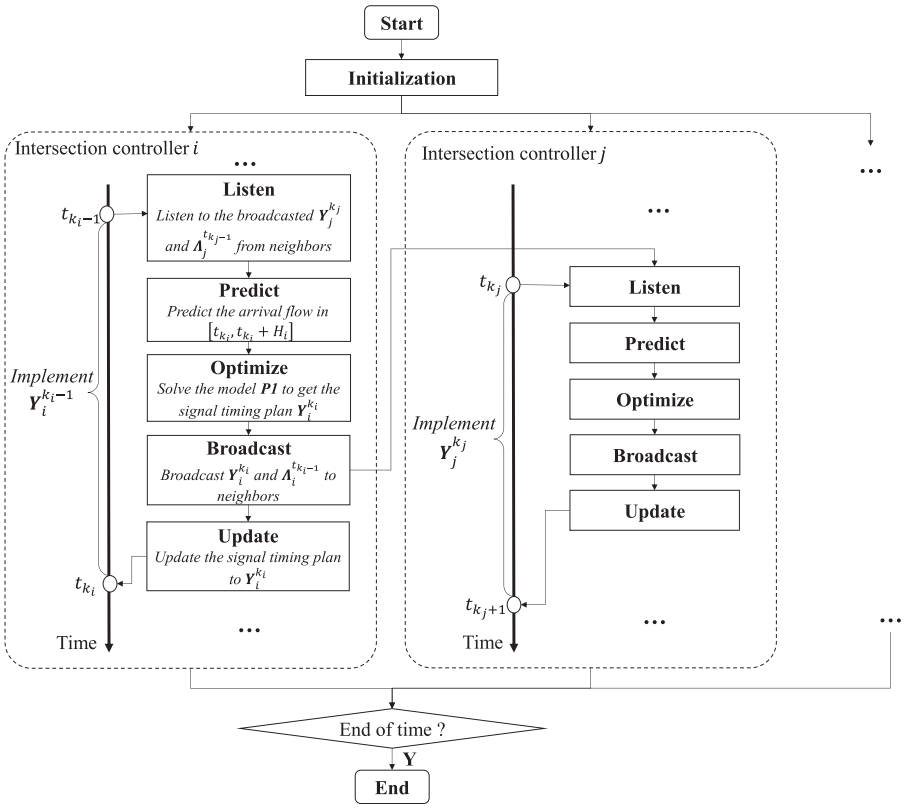

FIGURE 5 Parallel operation and rolling horizon scheme in the ADCSC framework.

3.3 | Individualized temporary parameters: horizon and update interval

The time latency from upstream departure flow to downstream creates room for the proposed ADCSC framework. This time latency (i.e., the travel time) determines the time required for upstream effects to propagate downstream. It allows downstream intersections to operate independently within a certain time interval. In this section, we discuss how varying travel times influence the optimization horizon and update interval.

Figure 6 illustrates two adjacent intersections: upstream intersection j and downstream intersection i. Intersection controller i will optimize the signal plan with a horizon of H_i every Δt_i. At each signal update time point, i needs the $\mathbf{Y}_j^{k_j}$ and $\mathbf{\Lambda}_j^{t_{k_j-1}}$ from j to start the optimization task.

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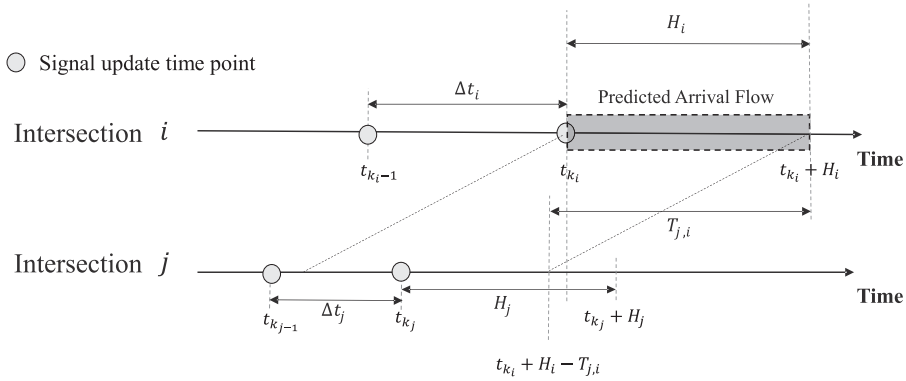

FIGURE 6 Design of optimization horizon and update interval.

Algorithm 1 The asynchronous framework of ADCSC.

Input: The detected queue length $\mathbf{\Lambda}_i^{t_{k_i}}$ of intersection $i \in \mathbf{I}$ in real time.

Output: Optimized signal plan $\mathbf{Y}_i^{k_i}$ for intersection $i \in \mathbf{I}$.

Initialization: Initialize the time step $t \leftarrow 0$. For each intersection $i \in \mathbf{I}$, initialize the horizon H_i and update interval Δt_i, the sequence number of signal plan $k_i \leftarrow 0$, the initial signal plan $\mathbf{Y}_i^{k_i}$, the signal update time point $t_{k_i} \leftarrow 0$, and the set of adjacent intersections $\mathbf{N}_i$.
In parallel, **for** each intersection $i \in \mathbf{I}$:
  if time step $t = k_i\,\Delta t_i$:
    implement the already optimized plan $\mathbf{Y}_i^{k_i}$
    listen to the broadcasted $\mathbf{Y}_j^{k_j}$ and $\mathbf{\Lambda}_j^{t_{k_j}}$ from $j \in \mathbf{N}_i$
    predict future arrival flow in $[t_{k_i+1}, t_{k_i+1} + H_i]$
    optimize model **P1** for i to obtain $\mathbf{Y}_i^{k_i+1}$
    update $\mathbf{Y}_i^{k_i+1}$ as the (k_i+1)th signal plan
    broadcast $\mathbf{Y}_i^{k_i+1}$ and $\mathbf{\Lambda}_i^{t_{k_i}}$ to intersections $j \in \mathbf{N}_i$
    $k_i \leftarrow k_i + 1$
  end if
end for
$t \leftarrow t + 1$
if t reaches the terminal time step:
  terminate the control framework
end if

Vehicles from upstream intersection j will arrive at downstream intersection i after travel time $T_{j,i}$. This study adopts the point-queue model to calculate the travel time (Ban et al., 2012). It assumes that traffic flow travels at the free flow speed for the entire length of the link until it gets to the stop bar. Although the physical length of each vehicle is ignored, the point-queue model can still preciously capture spatial and temporal traffic state (X. Wang et al., 2024). It has been widely used in many signal timing optimization models (H. Liu & Gayah., 2024; H. Yu et al., 2018; Xu et al., 2024).

Intersection i can use the previous information from j ($\mathbf{Y}_j^{k_j}$ and $\mathbf{\Lambda}_j^{t_{k_j-1}}$) to predict future arrival flow and start the optimization. To ensure i can start the optimization, the received $\mathbf{Y}_j^{k_j}$ should be sufficient to predict arrival flow. If H_i and Δt_i are too long, the prediction may require the next signal plan $\mathbf{Y}_j^{k_j+1}$, which are not ready for j. Therefore, H_i and Δt_i are crucial parameters to ensure the effective operation of the asynchronous framework.

The arrival flow at i at $t_{k_i} + H_i$ is the depart flow at j at $t_{k_i} + H_i - T_{j,i}$. Therefore, intersection controller i should be aware of the signal plan of j at $t_{k_i} + H_i - T_{j,i}$ (i.e., it should be included in $\mathbf{Y}_j^{k_j}$). If Equation (1) always holds, the received $\mathbf{Y}_j^{k_j}$ are sufficient for i to start the optimization:

$$t_{k_i} + H_i - T_{j,i} \le t_{k_j} + H_j \tag{1}$$

To facilitate subsequent calculations and analysis, Equation (2) is required to be satisfied in this study (as shown in Figure 6):

$$t_{k_i} + H_i - T_{j,i} \le t_{k_i} \le t_{k_j} + H_j \tag{2}$$

Then Equations (3) and (4) can be derived:

$$H_i \le T_{j,i} \tag{3}$$

$$t_{k_i} - t_{k_j} \le H_j \tag{4}$$

If Equations (3) and (4) hold, then Equation (1) will naturally hold and the i can obtain enough information to start the optimization.

Further, due to $t_{k_i-1} \le t_{k_j}$ and $t_{k_i} - t_{k_i-1} = \Delta t_i$, it always has $t_{k_i} - t_{k_j} \le \Delta t_i$. Therefore, if $\Delta t_i \le H_j$, the Equation (4) will be always satisfied:

$$t_{k_i} - t_{k_j} \le \Delta t_i \le H_j \tag{5}$$

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Thus, Equations (3) and (5) limit the range of H_i and Δt_i.

Besides, the Δt_i should be always smaller than the local optimization horizon H_i:

$$\Delta t_i \leq H_i \tag{6}$$

In summary, the optimization horizon H_i should not exceed the free flow travel time (as shown in Equation 3), and the update interval Δt_i should not exceed the optimization horizons of both i and j (as shown in Equations 5 and 6).

In the road network, the intersection i can have many upstream intersections. To ensure the asynchronous framework can operate effectively in arbitrary networks, the values of H_i and Δt_i are determined by Equations (7) and (8):

$$H_i \leq \min\left\{T_{j,i} | \forall j \in \mathbf{N}_i\right\},\ \forall i \in \mathbf{I} \tag{7}$$

$$\Delta t_i \leq \min\left\{\min\left\{H_j | \forall j \in \mathbf{N}_i\right\}, H_i\right\},\ \forall i \in \mathbf{I} \tag{8}$$

Equations (7) and (8) show that the optimization horizon and update interval will vary for different intersections when the travel time $T_{j,i}$ of each link differs. For simplification, this study sets $H_i = \min\{T_{j,i}\}$ and $\Delta t_i \leq \min\{H_j, H_i\}$ in this study. Then, the inequality (1) will always be satisfied, and the i can obtain enough information from adjacent intersections to predict future arrival flow and start the optimization.

4 | SIGNAL TIMING PLAN OPTIMIZATION MODEL

This section explains the formulation of the MILP model for signal timing plan optimization problem **P1** for a certain intersection i at a signal update time point t_{k_i} within the asynchronous decentralized framework. The optimization problem for each intersection is formulated in the same way. Without loss of generality, we formulate the k_ith signal optimization problem of intersection i as an example in this section. The main contribution of this study is to propose a general asynchronous framework for multi-agents decision-making problems, ensuring that each agent receives sufficient input information to optimize local plans independently. Any proactive DCSC problem formulation can benefit from this framework.

4.1 | Objective and decision variables

The objective function of signal timing optimization for a certain intersection i is to minimize the cumulative queue length at each time step $\lambda_{i,m}(t)$ in all the approach lanes during the horizon H_i, which is helpful to reduce vehicle waiting time and delay. The objective function of **P1** is:

$$\min_{\mathbf{Y}_i^{k_i}} \sum_{t=t_{k_i}}^{t_{k_i}+H_i} \sum_{m \in \mathbf{M}_i} \lambda_{i,m}(t) \tag{9}$$

The decision variable is the signal timing plan $\mathbf{Y}_i^{k_i}$:

$$Y_i^{k_i} = \left\{\left[t_{i,m}^{k_i,\mathrm{s}}, t_{i,m}^{k_i,\mathrm{e}}\right]\right\},\ \forall m \in \mathrm{M}_i \tag{10}$$

where $t_{i,m}^{k_i,\mathrm{s}}$ and $t_{i,m}^{k_i,\mathrm{e}}$ represent the start and end time of the green light of movement m at intersection i in the k_ith signal plan duration, respectively. In the optimization model, cycles are not explicitly considered. A more flexible signal scheme without fixed phase structure and sequence is generated.

4.2 | Signal timing constraints

For a certain movement m at intersection i, each continuous green duration should fall between the minimum green time $g_{\min}$ and the maximum green time $g_{\max}$. Each movement will only be assigned green time at most once in the optimization model. Note that not every movement is guaranteed green time within the horizon; that is, $t_{i,m}^{k_i,\mathrm{s}}$ and $t_{i,m}^{k_i,\mathrm{e}}$ are not required to fall within the interval $[t_{k_i}, t_{k_i} + H_i]$.

$$t_{i,m}^{k_i,\mathrm{e}} \geq t_{i,m}^{k_i,\mathrm{s}} + g_{\min},\ \forall m \in \mathbf{M}_i \tag{11}$$

$$t_{i,m}^{k_i,\mathrm{e}} \leq t_{i,m}^{k_i,\mathrm{s}} + g_{\max},\ \forall m \in \mathbf{M}_i \tag{12}$$

The green lights in different phases should satisfy constraints in Equations (13) and (14): For two conflicting movements m and n (e.g., the northbound and eastbound through movements), this study introduces $\pi_{m,n}$ to indicate the safety time gap between their green intervals. Note that the value of $\pi_{m,n}$ can be decided dynamically according to the real-time traffic condition (Long et al., 2020). This study just takes a fixed value in a conservative way for simplification.

$$t_{i,n}^{k_i,\mathrm{s}} \geq t_{i,m}^{k_i,\mathrm{e}} + \pi_{m,n} - \mathrm{M}\left(2 - \gamma_{m,n} - \varphi_{m,n}^{k_i}\right) \quad \forall m, n \in \mathbf{M}_i \tag{13}$$

$$t_{i,n}^{k_i,\mathrm{e}} \leq t_{i,m}^{k_i,\mathrm{s}} - \pi_{m,n} + \mathrm{M}\left(1 - \gamma_{m,n} + \varphi_{m,n}^{k_i}\right) \quad \forall m, n \in \mathbf{M}_i \tag{14}$$

where M is a sufficiently big number. $\gamma_{m,n}$ is the binary parameter representing the incompatibility of movements

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m and n. $\gamma_{m,n} = 1$, if movements m and n are incompatible; $\gamma_{m,n} = 0$, otherwise. $\varphi_{m,n}^{k_i}$ is the binary variable representing sequence of movements m and n. $\varphi_{m,n}^{k_i} = 1$, if the phase of movement n is after m in the k_ith signal plan; $\varphi_{m,n}^{k_i} = 0$, otherwise.

4.3 | Smooth switch constraints

The signal plan is updated frequently due to the rolling horizon scheme. To avoid the sudden change of signal plans, some constraints need to be proposed to ensure a smooth switch between two adjacently updated plans (C. Ma et al., 2023).

To keep the continuity of green time between two adjacent signal plans, if the movement m at intersection i is allocated green time in the $(k_i - 1)$th signal plan and the green duration will continue to the next optimization horizon (i.e., $t_{i,m}^{k_i-1,\mathrm{s}} \le t_{k_i}$ *and* $t_{i,m}^{k_i-1,\mathrm{e}} \ge t_{k_i}$), then Equation (15) is proposed to continue the green duration:

$$t_{i,m}^{k_i,\mathrm{s}} = t_{i,m}^{k_i-1,\mathrm{s}} \mid \left(t_{i,m}^{k_i-1,\mathrm{s}} \le t_{k_i} \text{ and } t_{i,m}^{k_i-1,\mathrm{e}} \ge t_{k_i}\right) \quad \forall m \in \mathbf{M}_i \tag{15}$$

From Equation (15), the start time point from the $(k_i - 1)$th signal plan is reserved for the k_ith signal plan to keep the continuity of green time.

Additionally, to avoid the insufficient safety time gap, it needs to reserve the red time to ensure sufficient safety time gap between two conflicting movements m and n between the $(k_i - 1)$th and k_ith signal plans. If movement n is assigned the green light in the last optimization (i.e., $t_{i,m}^{k_i-1,\mathrm{e}} \le t_{k_i}$), then Equation (16) should be satisfied:

$$t_{i,n}^{k_i,\mathrm{s}} \ge t_{i,m}^{k_i-1,\mathrm{e}} + \pi_{m,n} - \mathrm{M}\left(1 - \gamma_{m,n}\right) \mid t_{i,m}^{k_i-1,\mathrm{e}} \le t_{k_i} \quad \forall m, n \in \mathbf{M}_i \tag{16}$$

4.4 | Queue length prediction constraints

The evolution of the queue length can be derived using conservation Equation (17):

$$\lambda_{i,m}(t) = \lambda_{i,m}(t-1) + q_{i,m}^{\mathrm{in}}(t) - q_{i,m}^{\mathrm{out}}(t), \quad \forall m \in \mathbf{M}_i, \forall t = t_{k_i-1}+1, \dots, t_{k_i}, \dots, t_{k_i}+H_i \tag{17}$$

where $\lambda_{i,m}(t)$ denotes the queue length (represented by stopped vehicle numbers at the stop line) of movement m at intersection i at time t. $q_{i,m}^{\mathrm{in}}(t)$ and $q_{i,m}^{\mathrm{out}}(t)$ are numbers of arriving vehicles and discharging vehicles of movement m at intersection i at time t, respectively. In the rolling horizon scheme, at the signal update time point t_{k_i-1}, the task is to optimize the signal plan during $[t_{k_i}, t_{k_i} + H_i]$. Therefore, the queue dynamics during $[t_{k_i-1}, t_{k_i} + H_i]$ should be predicted.

The arrival flow $q_{i,m}^{\mathrm{in}}(t)$ during $[t_{k_i-1}, t_{k_i} + H_i]$ can be predicted by:

$$q_{i,m}^{\mathrm{in}}(t) = \sum_j \sum_n p_{n,m} \cdot \min\left\{\begin{matrix} \delta_{j,n}\left(t - T_{j,i}\right) \cdot s_{j,n} \cdot l_{j,n} \cdot t_s, \\ \lambda_{j,n}\left(t - T_{j,i}\right) \end{matrix}\right\}, \quad \forall m \in \mathbf{M}_i \cap \mathbf{M}_{CL}, \forall j \in \mathbf{N}_i, \forall n \in \mathbf{U}_m \cap \mathbf{M}_j, \quad \forall t = t_{k_i-1}+1, \dots, t_{k_i}, \dots, t_{k_i}+H_i \tag{18}$$

where $\mathbf{U}_m$ denotes the set of upstream movements of m. $p_{n,m}$ is the turning rate of the direction from n to m. $\delta_{i,m}(t)$ is a binary variable that denotes the signal state for movement m at intersection i at time t ($\delta_{i,m}(t) = 1$, if the signal for movement m at intersection i at time t is green; $\delta_{i,m}(t) = 0$, otherwise). $\mathbf{M}_{CL}$ denotes the set of movements on connecting links. The connecting link refers to the link in the network with an upstream interaction. $s_{i,m}$ denotes the saturation flow. t_s is the duration of a time step. $l_{i,m}$ is the total number of lanes dedicated to movement m at intersection i.

In Equation (18), the first term in the min{·} operator is the saturation flow of upstream at time $t - T_{j,i}$. The second term is the queue length of upstream at time $t - T_{j,i}$. The actual traffic flow into the downstream movement m is limited by these two terms. The signal state $\delta_{j,n}(t - T_{j,i})$ and queue length $\lambda_{j,n}(t - T_{j,i})$ are known parameters in the received $\boldsymbol{Y}_j^{k_j}$ and $\boldsymbol{\Lambda}_j^{t_{k_j}}$.

If the movement m at intersection i is on the source link, which refers to the incoming link at the boundary of the network without upstream intersections, the $q_{i,m}^{\mathrm{in}}(t)$ is determined by the arrival flow rate $a_{i,m}$:

$$q_{i,m}^{\mathrm{in}}(t) = a_{i,m}, \quad \forall m \in \mathbf{M}_i \cap \mathbf{M}_{SL}, \quad \forall t = t_{k_i-1}+1, \dots, t_{k_i}, \dots, t_{k_i}+H_i \tag{19}$$

where the $\mathbf{M}_{SL}$ denotes the set of source links. Vehicles are assumed to be generated at source links only in this study. Through Equations (18) and (19), $q_{i,m}^{\mathrm{in}}(t)$ is an already predicted parameter in the model. The $q_{i,m}^{\mathrm{out}}(t)$ is determined by:

$$q_{i,m}^{\mathrm{out}}(t) = \min\left\{\delta_{i,m}(t) \cdot s_{i,m} \cdot l_{i,m} \cdot t_s, \lambda_{i,m}(t-1)\right\}, \quad \forall m \in \mathbf{M}_i, \forall t = t_{k_i-1}+1, \dots, t_{k_i}, \dots, t_{k_i}+H_i \tag{20}$$

where $\delta_{i,m}(t)$ is a variable that is determined by $t_{i,m}^{k_i,\mathrm{s}}$ and $t_{i,m}^{k_i,\mathrm{e}}$ during the optimization horizon $[t_{k_i}, t_{k_i} + H_i]$:

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$$t + \mathrm{M}\left[1 - \delta_{i,m}(t)\right] \geq t_{i,m}^{k_i,\mathrm{s}}, \quad \forall m \in \mathbf{M}_i,\ \forall t = t_{k_i}, \ldots, t_{k_i} + H_i \tag{21}$$

$$t - \mathrm{M}\left[1 - \delta_{i,m}(t)\right] \leq t_{i,m}^{k_i,\mathrm{e}}, \quad \forall m \in \mathbf{M}_i, \forall t = t_{k_i}, \ldots, t_{k_i} + H_i \tag{22}$$

Therefore, $q_{i,m}^{\mathrm{out}}(t)$ during the optimization horizon $[t_{k_i}, t_{k_i} + H_i]$ is a decision variable corresponding to $\mathbf{Y}_i^{k_i}$.

It is also important to limit the outflow vehicles should not exceed the maximum capacity at downstream links:

$$p_{m,n} \cdot q_{i,m}^{\mathrm{out}}(t) \leq N_{j,n} \cdot l_{j,n}, \quad \forall m \in \mathbf{M}_i,\ \forall j \in \mathbf{N}_i, \forall n \in \mathbf{M}_j \cap \mathbf{D}_m, \quad \forall t = t_{k_i}, \ldots, t_{k_i} + H_i \tag{23}$$

where $\mathbf{D}_m$ is the set of all closely downstream movements of movement m. And $n \in \mathbf{M}_j \cap \mathbf{D}_m$ is the downstream movement of movement m at intersection j. $N_{j,n}$ is the maximum storage of an approach lane for the movement n at intersection j, and $l_{j,n}$ is the approach lane number of the movement n at intersection j. $N_{j,n} \cdot l_{j,n}$ denotes the total available storage for the movement n at intersection j. The point-queue model assumes that each link can store an infinite queue, which may lead to concerns about spillback or gridlock. Equation (23) enhances the point-queue model by accounting for the physical queue storage and ensuring it does not exceed the available capacity. It is helpful to prevent queue spillback in the road network.

4.5 | Constraint linearization

The formulation of Equation (20) is nonlinear and makes the mathematical program hard to solve. To avoid the complexity to solve the non-linear problem, this study converts it to Equations (20.1)–(20.2) as follows:

$$q_{i,m}^{\mathrm{out}}(t) \leq \delta_{i,m}(t) \cdot s_{i,m} \cdot l_{i,m} \cdot t_s, \quad \forall m \in \mathbf{M}_i \forall t = t_{k_i}, \ldots, t_{k_i} + H_i \tag{20.1}$$

$$q_{i,m}^{\mathrm{out}}(t) \leq \lambda_{i,m}(t-1), \quad \forall m \in \mathbf{M}_i,\ \forall t = t_{k_i}, \ldots, t_{k_i} + H_i \tag{20.2}$$

The Equations (20.1) and (20.2) limit $q_{i,m}^{\mathrm{out}}(t)$ to certain values during $[t_{k_i}, t_{k_i} + H_i]$; however, the optimal solution may not be binding on any of them (Doan & Ukkusuri, 2012). To this end, the $q_{i,m}^{\mathrm{out}}(t)$ should be maximized in the objective function. Therefore, Equation (9) should be rewritten as Equation (24) to maximize the total throughput while minimizing the total queue length during $[t_{k_i}, t_{k_i} + H_i]$.

$$\max \sum_{t=t_{k_i}}^{t_{k_i}+H_i} \sum_{m \in \mathbf{M}_i} q_{i,m}^{out}(t) - \sum_{t=t_{k_i}}^{t_{k_i}+H_i} \sum_{m \in \mathbf{M}_i} \lambda_{i,m}(t) \tag{24}$$

The $\lambda_{i,m}(t)$ is a monotonous decrease function of $q_{i,m}^{out}(t)$ (as shown in Equation 17). Therefore, objective (24) is a monotonous increase function of $q_{i,m}^{out}(t)$. To maximize the objective, $q_{i,m}^{out}(t)$ will be naturally bound to the upper boundary of the feasible region, which is limited by Equations (20.1)–(20.2). In this way, $q_{i,m}^{out}(t)$ will always be the minimum value of two boundary values. Therefore, the new formulations (20.1)–(20.2) are equivalent to the initial formulation (20).

Equation (24) and Equations (11)–(19), (20.1)–(20.2), and (21)–(23) formulate the optimization problem **P1** in the proposed ADCSC framework for each intersection. The **P1** is a MILP model and can be conveniently solved by the branch-and-bound method in commercial solvers such as Gurobi or CPLEX.

5 | SIMULATION EXPERIMENTS

The proposed ADCSC framework decouples the interdependent relationship between demand prediction and decision-making across intersections within the network through an asynchronous approach, achieving more accurate demand forecasting in the proactive control. This chapter first assesses the performance of the proposed method through traffic simulation tests on a simulated road network, comparing traffic performance against the other coordinated control strategies. A sensitivity analysis is conducted on the temporary parameters within the asynchronous framework. Furthermore, the simulation experiment is conducted on a real-world road network to evaluate the proposed ADCSC strategy. The code of the proposed ADCSC framework in a demo simulated road network is available at: https://github.com/bryan-zjc/ADCSC-framework.

5.1 | Simulation settings and benchmarks

The road network topology in the simulation tests is illustrated in Figure 7. It adopts an asymmetric design with intersection distances varying from 300 to 500 m to enhance generality. Each intersection controller has an individual H_i and Δt_i in the ADCSC framework due to different lengths of links. The layout of each intersection contains four arms, each equipped with three lanes

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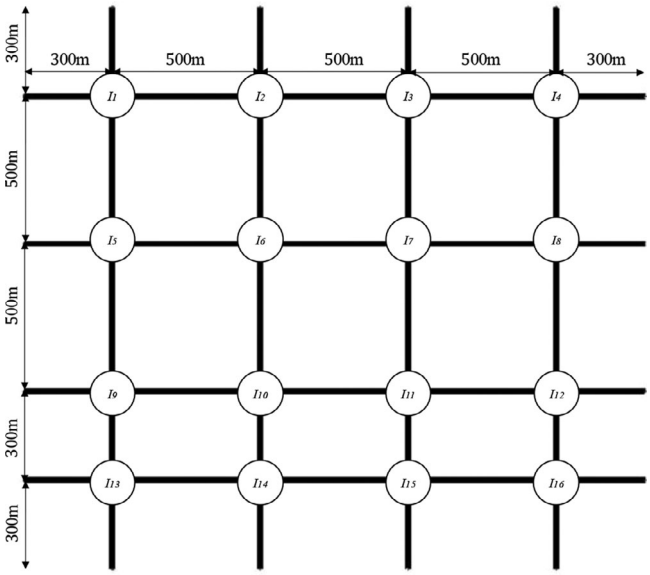

FIGURE 7 The simulated road network for evaluation.

dedicated to left turn, through, and right turn, respectively. The right-turning movement is served by a dedicated lane at each intersection. Therefore, right-turning vehicles do not have conflicts with other vehicles, and they will not be controlled by the signal in this study. The traffic light for right-turning movement remains green all the time. Note that our proposed model is still applicable in the case that the traffic light should control the right-turning movement.

1. Symmetric low demand (SLD): 400 vehicles per hour in each source link.
2. Symmetric high demand (SHD): 800 vehicles per hour in each source link.
3. Asymmetric low demand (ALD): 600 vehicles per hour in east and west source links and 200 vehicles per hour in north-south source links.
4. Asymmetric high demand (AHD): 1200 vehicles per hour in east and west source links and 600 vehicles per hour in north-south source links.

Table 3 presents the primary parameters of the proposed ADCSC framework. Signal timing parameters follow Koonce (2008), and the saturation flow rate is according to Al Islam and Hajbabaie (2017).

Experiments are conducted in Simulation of Urban Mobility (Krajzewicz et al., 2012), which is a widely recognized platform for traffic simulation (X. Wang et al., 2024). The traffic signal is controlled by the *TraCI* interface in Python. The problem **P1** for each intersection is solved by Gurobi 9.1.2 on a Windows server with an Intel 4.20 GHz CPU and 32 GB of RAM. All experiments run for 1800 s after a warm-up period of 500 s to ensure the simulated road network achieves a stable traffic condition.

TABLE 3 Basic information of the simulation.

Parameters	Value
Free flow speed (m/s)	11.11 (40 km/h)
Speed limit (m/s)	16.67 (60 km/h)
t_s (s)	1
$s_{i,m}$ (veh/h/lane)	1800
$\pi_{m,n}$ (s)	3
g_{max} (s)	50
$g_{\min}$ (s)	10

In our experiment, the performance of the proposed ADCSC strategy is evaluated in comparison with three benchmark strategies:

1. MULTIBAND-96 strategy: The MULTIBAND-96 model (Stamatiadis & Gartner, 1996) is a centralized method that optimizes signal variables (e.g., phase lengths, offsets, and cycles) to generate variable bandwidth progressions. It compromises the system coordination performance by only providing bandwidth in the main direction (the east-west direction in the simulated network).
2. Cyclic back pressure (CBP) strategy in Le et al. (2015): In the CBP strategy, each controller adjusts the duration of the green light based on the number of queued vehicles at the start of each cycle, without incorporating traffic evolution prediction.
3. Al Islam et al. (2020) strategy: It is a synchronous DCSC framework. In this strategy, all controllers have the same optimization horizon and update interval. Some controllers may receive insufficient information to predict future arrival flow. It assumes that the last optimized signal plan stays unchanged when predicting arrival flow.

5.2 | Traffic performance

Five measurements are adopted to evaluate the traffic performance under different strategies: average delay (per vehicle per intersection), average vehicle travel speed (per vehicle), average number of stops (per vehicle per intersection), and average fuel consumption efficiency (per vehicle). The test results of each control strategy under all demand conditions are shown in Table 4. The emission model in the report of the US Environmental Protection Agency is applied for the estimation of average vehicle fuel consumption efficiency (EPA, 2002). This model first calculates vehicle-specific power based

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TABLE 4 Network performance for different signal timing control strategies in the simulated network.

Demand condition	Control strategy	Performance measures			
		Average delay (s/veh per intersection)	Average travel speed (km/h)	Average number of stops (per vehicle per intersection)	Average fuel consumption efficiency (mL/km/veh)
SLD	MULTIBAND-96	47.18	17.03	0.86	141.83
	Le et al. (2015)	23.15	25.40	0.66	71.33
	Al Islam et al. (2020)	16.69	29.39	0.49	55.13
	ADCSC	15.94	29.93	0.44	45.89
SHD	MULTIBAND-96	71.69	12.58	1.28	250.67
	Le et al. (2015)	71.77	12.70	1.67	224.44
	Al Islam et al. (2020)	47.70	16.85	1.04	136.25
	ADCSC	34.58	20.55	0.72	94.21
ALD	MULTIBAND-96	33.97	20.70	0.64	96.69
	Le et al. (2015)	22.89	25.53	0.62	72.31
	Al Islam et al. (2020)	14.37	31.13	0.46	49.47
	ADCSC	14.16	31.30	0.41	42.59
AHD	MULTIBAND-96	77.52	11.89	1.39	278.76
	Le et al. (2015)	73.91	12.39	1.93	263.45
	Al Islam et al. (2020)	60.56	14.35	1.38	206.06
	ADCSC	46.89	17.05	0.93	135.74

Abbreviations: ADCSC, asynchronous decentralized coordinated signal control; AHD, asymmetric high demand; ALD, asymmetric low demand; SHD, symmetric high demand; SLD, symmetric low demand.

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on vehicle speed and acceleration. Then, the vehicle fuel consumption is determined through a look-up table. Additionally, vehicle types, engine sizes, and mileages will also affect the calculation of fuel consumption. In this study, the coefficients for vehicle Category 11 in the model (odometer < 50,000 miles and engine size < 3.5 L) are applied.

As shown in Table 4, all three decentralized approaches outperform the centralized MULTIBAND-96 strategy. MULTIBAND-96 focuses on the main arterial's bandwidth progression (east-west direction in this study), often at the expense of the north-south direction. As a result, our ADCSC approach achieves the greatest delay reduction in symmetric demand conditions (66.21% in SLD and 51.76% in SHD), compared to it. MULTIBAND-96 can effectively reduce the average number of stops, particularly under high demand. It can ensure vehicles will experience fewer stops than CBP strategy in ALD and AHD, by providing smoother progression in the main direction.

The CBP strategy in Le et al. (2015) is without the traffic evolution prediction. It only uses the detected queue length at the beginning of each cycle. When traffic demand is high (SHD and AHD), CBP even leads to severe congestion, while the ADCSC framework can still provide efficient signal plans to coordinate traffic flows from different directions. In the proposed ADCSC framework, future traffic demand is predicted using information from adjacent intersections. It realizes the proactive prediction in the decentralized approach. The proposed ADCSC strategy can reduce the average delay by 51.82% and 36.56% in SHD and AHD, respectively. Besides, the ADCSC strategy can also provide smooth vehicle trajectories with fewer stop counts, further can improve vehicle speed (61.81% in SHD and 37.61% in AHD) and reduce fuel consumption (58.02% in SHD and 48.48% in AHD) significantly.

Al Islam et al. (2020) strategy is a decentralized approach that can predict future traffic evolution. However, due to the complex coupling relationship between state prediction and signal optimization, it predicts arrival flow from the unchanging signal plan. It may generate good signal plans in low-demand conditions (e.g., SLD and ALD) because the arrival flow in low-demand conditions fluctuates within a narrow range, and the predicted arrival flow might be similar to the real arrival pattern. However, accurate prediction of traffic flow evolution is of paramount importance to generate efficient signal plans, especially in high-demand conditions (Ghosh-Dastidar & Adeli, 2006; Jiang & Adeli, 2004b). The ADCSC framework decouple and split the prediction and optimization, providing more accurate traffic evolution for signal optimization. In SHD and AHD, the ADCSC strategy reduces the average delay (27.51% in SHD and 22.57% in AHD), average stop counts (30.77% in SHD and 32.61% in AHD), and fuel consumption (30.86% in SHD and 34.13% in AHD) significantly.

Figure 8 illustrates the spatiotemporal trajectories of vehicles on the network route passing by intersections (I_5, I_6, I_7, I_3) under SLD and ALD, respectively. Here, we compare three decentralized approaches (Al Islam et al., 2020; Le et al., 2015; and our ADCSC strategy) because the MULTIBAND-96 has been shown to be inefficient for coordinating network flow. The horizontal axis represents time, while the vertical axis shows the cumulative distance along the route. Horizontal bars depict the signal timing at corresponding intersections. SLD is the symmetric demand condition with evenly distributed demand in all directions, while in ALD, the demand in the west-east direction is significantly higher than the demand in the north-west.

As shown in Figure 8a,b, CBP allocates more green time to the direction with higher queue length at the beginning of each cycle, and therefore the generated signal plans are unable to adapt to future demand changes. Vehicles arriving at the green light are prone to stopping and waiting, particularly under asymmetric demand conditions, where the direction with higher demand may experience longer delays and a greater number of stops due to insufficient green duration.

The strategy proposed by Al Islam et al. (2020) is proactive and capable of optimizing the signal plan to adapt to future arrival flows. Consequently, as depicted in Figure 8c,d, this strategy can provide coordinated plans and ensure smoother vehicle trajectories, compared to CBP. However, the inherent prediction error of arrival flow may lead to unsuitable signal plans for the real demand. For instance, as illustrated in Figure 8c, the arrival flow is predicted based on the last optimized plans from adjacent intersections; however, adjacent intersections actually implement the newly optimized plans. This discrepancy results in an under-prediction of arrival flow, leading to long queues at I_6.

In contrast, Figure 8e,f shows that within the ADCSC framework, signal timing plans are efficiently coordinated in both symmetric and asymmetric demand conditions. The asynchronous approach enables more effective control performance, diminishing frequent stop-and-go and thereby reducing the associated safety risks, time cost, and energy consumption.

5.3 | Computational efficiency

The proposed ADCSC framework requires intersections within the network to efficiently solve their signal optimization models, ensuring timely updates to local signal

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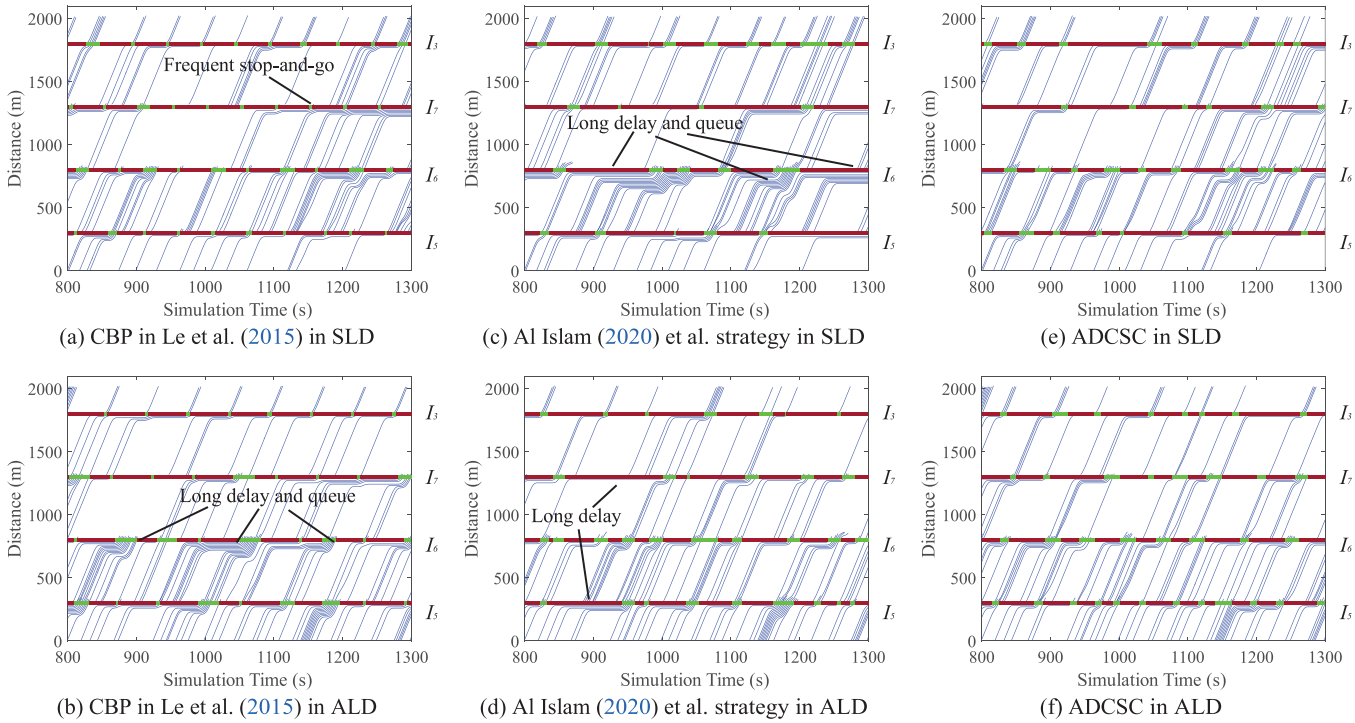

(a) CBP in Le et al. (2015) in SLD
(c) Al Islam (2020) et al. strategy in SLD
(e) ADCSC in SLD
(b) CBP in Le et al. (2015) in ALD
(d) Al Islam (2020) et al. strategy in ALD
(f) ADCSC in ALD

FIGURE 8 Trajectories on the route (I_5, I_6, I_7, I_3). (a) Cyclic back pressure (CBP) in Le et al. (2015) in symmetric low demand (SLD); (b) CBP in Le et al. (2015) in asymmetric low demand (ALD); (c) Al Islam et al. (2020) strategy in SLD; (d) Al Islam et al. (2020) strategy in ALD; (e) ADCSC in SLD; and (f) ADCSC in ALD.

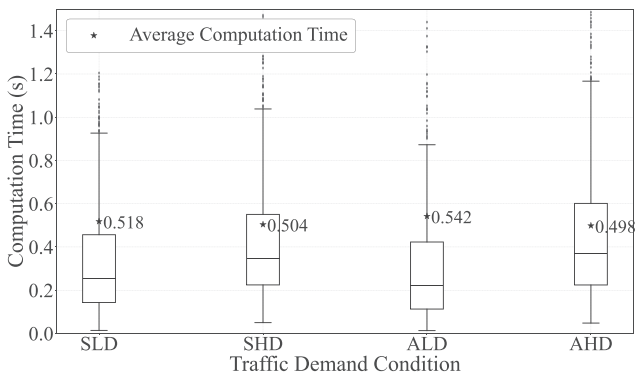

FIGURE 9 Model computation time of signal optimization model **P1** in different demand conditions.

timing and broadcasting the updates to neighboring intersections for signal coordination. As mentioned in Section 3.1, the computation time should not exceed the update interval Δt_i; otherwise, the controller will wait for optimization when the time step reaches to the signal update time point.

Figure 9 illustrates the average solution time for the signal optimization model across four demand conditions. The horizontal line in the box is the median of computation time, with error bars representing the standard deviations. The average computation time is marked as star scatter in the figure. It aggregates computation time across all intersections to show the average performance. In the simulated network, intersections I_6 and I_7 have longer link lengths, which result in larger optimization horizons and more complex signal problems. These larger problems require longer computation time, contributing to the outliers observed in the figure. Remarkably, the decentralized optimization model exhibits almost constant average computation time across varying demand scenarios. On average, the model **P1** can be effectively solved around 0.5 s across all demand profiles. Considering that the free flow travel time on a link is typically around 30 s, the update interval is tens of times longer than the computation time. It indicates the ADCSC framework can provide a time buffer for computation and communication in real-world application.

5.4 | Sensitive analysis

The optimization horizon H_i and update interval Δt_i is crucial parameters for the proposed ADCSC framework. As long as H_i and Δt_i satisfy Equations (7) and (8), the ADCSC framework can efficiently operate. However, a shorter H_i can reduce the model scale, and a longer Δt_i will reduce the number of signal updates and lead to a steadier signal plan. To this end, a sensitive analysis of H_i and Δt_i is performed to explore the potential benefits.

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5.4.1 | Optimization horizon

In the experiments above, for each intersection $i \in \mathbf{I}$, H_i is set to its maximum feasible value $\min\{T_{j,i}\}, \forall j \in \mathbf{N}_i$. This section compares the control performance and computational efficiency of the ADCSC framework with different optimization horizons H_i for the intersections. A reduction factor $\alpha \in \{0.5,\ 0.6, 0.7, 0.8, 0.9, 1\}$ is introduced to generate different feasible optimization horizons H_i^* in the sensitive analysis:

$$H_i^* = \alpha \cdot \min\left\{T_{j,i}\right\}\ , \forall j \in \mathbf{N}_i \tag{25}$$

The update interval varies with changes in the optimization horizon as shown in Equation (8). The impact of different update intervals will be further discussed in the next section. Here, we focus exclusively on the effect of varying optimization horizons.

Figure 10 shows the average delay and number of stops under different demand conditions and optimization horizons of the ADCSC framework. Under demand conditions SLD, SHD, and ALD, both average delay and stop counts initially increase and then decrease as α increases. The peak values occur around 0.8–0.9. This initial increase can be attributed to the growing prediction errors associated with a longer optimization horizon (Lin et al., 2023). Since the queue arrival flow prediction in Equation (18) is cumulative, the associated prediction errors also accumulate over time. However, when α exceeds 0.8 or 0.9, the average delay and stop counts begin to decrease. This is because the intersection controller can adapt to future traffic patterns more effectively over a longer optimization horizon. Consequently, the impact of errors is gradually smoothed out, leading to further improvements in system effectiveness.

In contrast, in demand condition AHD, both delay and number of stops exhibit a continuous increasing trend. This indicates that the point-queue model may generate significantly more errors under congested demand conditions. In directions with extremely high demand, these prediction errors are further amplified. Therefore, under the asymmetric high-demand condition, a smaller α is recommended to reduce prediction errors.

Figure 11 further shows the root mean square error of queue length prediction results $\lambda_{i,m}(t)$ in our ADCSC strategy under different optimization horizons. As the optimization horizon increases, these errors are compounded over time, leading to further amplification. It proves that the prediction error is larger under high-demand conditions (SHD and AHD), indicating that congestion will exacerbate the inaccuracies.

Figure 12 demonstrates the box plot of model computation time under different optimization horizons. The computation time is depicted as scatters, with error bars

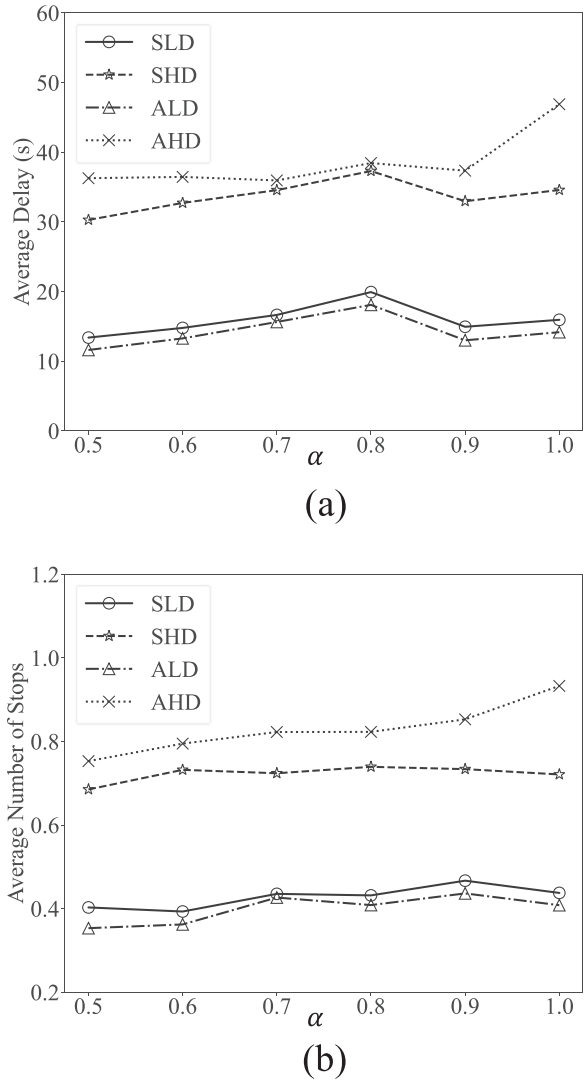

FIGURE 10 Network performance of ADCSC framework for different optimization horizons. (a) Average delay and (b) average number of stops.

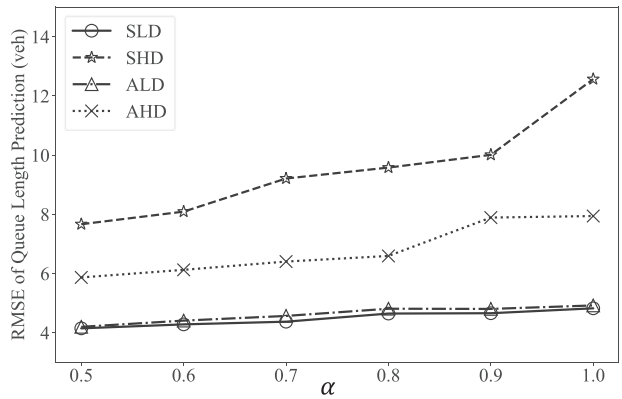

FIGURE 11 Root mean square error (RMSE) of the queue length in the ADCSC framework under different optimization horizons.

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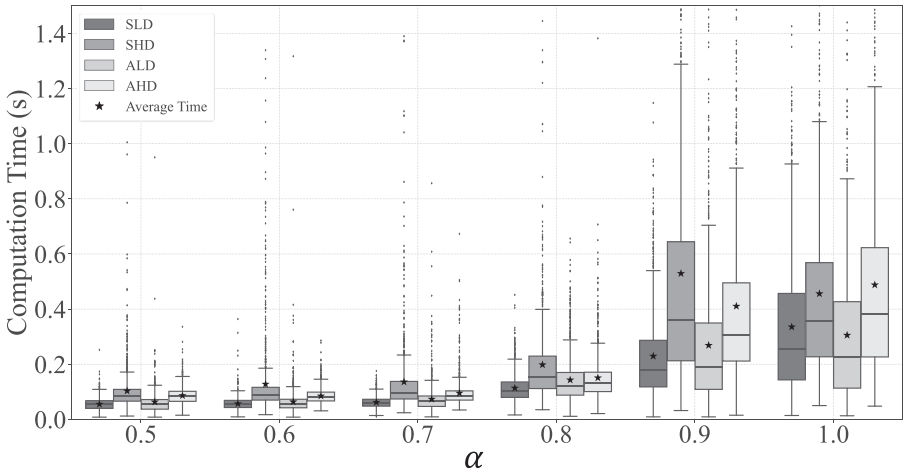

FIGURE 12 Computation time costs of ADCSC framework for different optimization horizons.

representing the standard deviations. As shown in the figure, the computation time of the ADCSC framework increases slightly with the optimization horizon. The optimization problem sizes increase with the optimization horizon, and therefore the ADCSC framework needs more computation time to predict traffic states and optimize signal plans. Meanwhile, the increasement of demand will also result in longer computational time. In real-world applications, a shorter optimization horizon (e.g., $\alpha = 0.5$) is recommended to provide better control performance and enhance computation efficiency.

5.4.2 | Update interval

If the value of the optimization horizon H_i decreases, the value of the update interval Δt_i may also become smaller, as Δt_i should be less than H_i. To analyze the impact of a decrease in Δt_i, this section compares the network performance and computation time under different update intervals. Another reduction factor $\beta \in \{0.5,\ 0.6, 0.7, 0.8, 0.9, 1\}$ is introduced:

$$\Delta t_i^* = \beta \cdot \min\left\{H_j, H_i\right\}\ ,\forall j \in \mathbf{N}_i \tag{26}$$

Figure 13 shows the control performance under different update intervals of the proposed ADCSC framework. The average delay stays stable with the increase of update interval in most conditions. The results prove that the average delay is not very sensitive to the update intervals. However, the average number of stops decreases with the update interval. Although Equations (15) and (16) are proposed to ensure the smooth switch, a shorter Δt_i^* will inevitably lead to more frequent changes in the signal plan. The green light may be frequently switched, and the stop-and-go number will increase (Hu et al., 2021). The long update

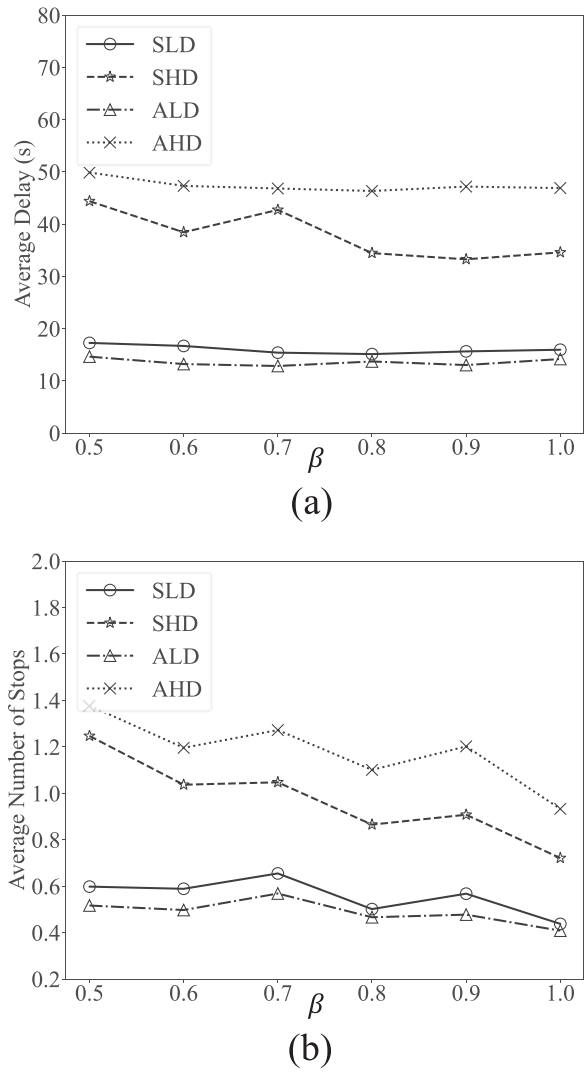

FIGURE 13 Network performance of ADCSC framework for different update intervals. (a) Average delay and (b) average number of stops.

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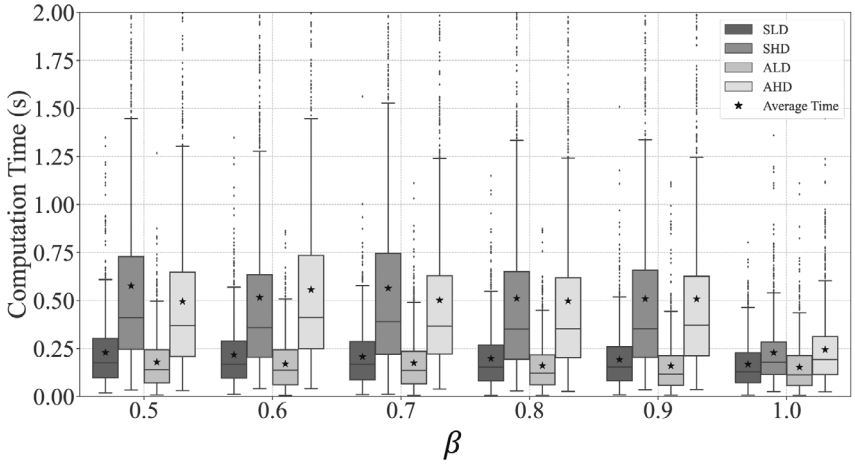

FIGURE 14 Computation time costs of ADCSC framework for different update intervals.

interval is beneficial to reducing the average number of stops.

The computation time of the ADCSC framework under different update intervals is shown in Figure 14. Due to the optimization H_i remains constant in each demand profile, the model scale remains essentially unchanged, and the average computation time stays almost constant under different values of β. However, it shows that the fluctuation and standard deviation of computation time decrease with the update interval. Since the optimization model needs to be frequently computed when the update interval is short, which leads to fluctuations in computation time and high standard deviation. With the increase of β, the computation time becomes more stable and exhibits less fluctuation, the possibility of long computation time will also be reduced. In summary, a longer update interval is recommended (e.g., $\beta = 1.0$) since it can decrease the number of stops and stabilize computation time without a significant increase in traffic delay.

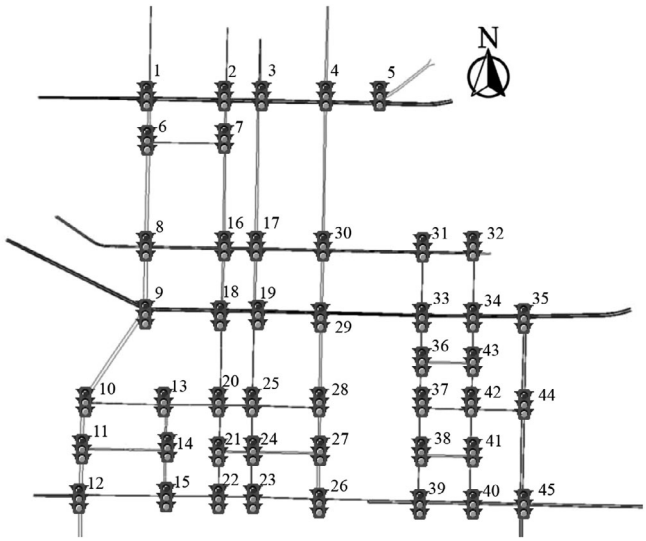

FIGURE 15 The real-world road network of Qinzhou City with 45 signalized intersections.

5.5 | Experiment in a real-world road network

To further evaluate the effectiveness and scalability of the proposed ADCSC framework, we further implement it into a real-world road network of Qinzhou City, China. This network includes 45 signalized intersections, both cross-shaped and T-shaped, as shown in Figure 15. The link lengths range from 210 to 673 m, and the simulation runs for 3600 s. The total traffic demand in this network was collected from 7:00 to 8:00 a.m. on December 14, 2023, with 16,236, 22,088, and 17,670 veh/h during the intervals 7:00–7:20 a.m., 7:20–7:40 a.m., and 7:40–8:00 a.m., respectively. We evaluate three decentralized approaches (CBP strategy in Le et al., 2015; Al Islam et al., 2020, strategy, and the proposed ADCSC) in this real-world network.

The average delay per intersection for three decentralized methods was evaluated. The CBP strategy in Le et al. (2015) had the highest delay at 62.86 s/veh, while the Al Islam et al. (2020) strategy showed 38.33 s/veh. The proposed ADCSC framework achieved the best performance with the lowest delay of 33.61 s/veh. Specifically, the ADCSC approach reduces average delay by 46.53%, compared to the CBP strategy from Le et al. (2015) and by 12.31%, compared to the strategy from Al Islam et al. (2020). This indicates that our asynchronous method effectively coordinates intersections across the real-world road network with varying intersection types and link lengths.

Figure 16 further shows the computation time of the ADCSC framework in the real-world network. The results prove that our decentralized asynchronous framework can solve the signal optimization model efficiently for each intersection, even in a larger road network. In most scenarios, each intersection can generate an optimal signal

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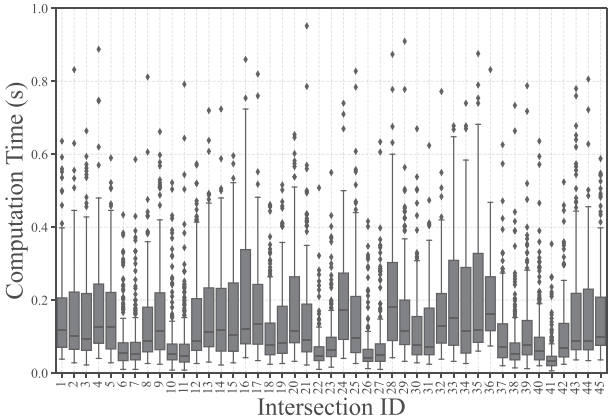

FIGURE 16 Computation time of ADCSC framework in the real-world network.

plan within 0.8 s. The ADCSC framework is scalable across different network sizes without computational efficiency degradation. The results demonstrate the scalability of the ADCSC framework in large real-world road networks, with each intersection controller solving the local signal problem less than 1 s.

6 | CONCLUSION AND FUTURE WORK

This study introduces an asynchronous framework for decentralized proactive coordinated signal control for the intersections in the urban road network. This approach decouples the entangled relationship between decision-making and state prediction among different agents in decentralized coordinated decision-making problems, enabling agents to proceed with collaborative decisions without waiting for other agents' optimization outcomes. The proposed ADCSC framework enables each intersection to predict the future arrival flow and optimizes its own signal timing in an asynchronous way based on the previous signal plans and traffic states of neighbors. A rolling horizon scheme is used to accommodate fluctuating traffic conditions and provide a time buffer for computation time. A flexible signal scheme without a fixed phase structure and sequence is adopted for signal timing optimization to better adapt to traffic demand.

The simulation results have demonstrated that the proposed ADCSC framework is capable of adapting to varying traffic conditions in real time. It provides more efficient network-wide signal coordination, resulting in higher travel speeds and fewer stops compared to the benchmark signal coordination methods. The model can be solved in real time within a relatively short time, and the proposed ADCSC framework can provide plenty of time for model computation. Sensitive analysis about the values of two temporary parameters is conducted, providing guidance for parameter value in application. We further evaluate the effectiveness and scalability of the ADCSC framework in a real-world network with 45 signalized intersections.

The ADCSC framework can be smoothly integrated into urban road networks using existing infrastructure, such as sensors, communication devices, and control modules. Its decentralized design ensures scalability to networks of any size, making it ideal for large urban areas. With a longer update interval, compared to its computation time, the framework offers high computational efficiency in real-world applications. These advantages make ADCSC a practical and effective solution for future urban traffic management.

For future research, real-world validation is essential to solidify the contributions of the proposed method. In real-world scenarios, many potential challenges may arise within the ADCSC framework. First, ensuring the accuracy and reliability of traffic data is crucial in the ADCSC framework. Future work should focus on developing advanced data processing algorithms capable of filtering noise, correcting errors, and integrating data from multiple sources. Second, future research can develop advanced queue models that account for heterogeneous travel times and turning behavior, thereby enhancing the ADCSC framework's performance. Third, when applied in real-world scenarios, the ADCSC framework must be capable of handling communication delays, interruptions, or losses. To enhance robustness, it is necessary to develop strategies for mitigating the impacts of these communication issues, such as by implementing adaptive control strategies that can maintain system stability under these adverse conditions. Furthermore, integrating the proposed ADCSC framework at the intersection level with macroscopic traffic control at the traffic zone level, such as perimeter control, could further improve traffic efficiency in large, city-wide road networks.

ACKNOWLEDGMENTS

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